

Ph429 NLD & Chaos Fall 2026

Problem Set One

Some of these problems are from Strogatz Ch.10 or are strongly inspired by those. I had hope we would be able to study up to and including section 5 (Lyapunov exponent) but it looks like we'll need to stop at section 4 for now, but we'll return to sections 5 and parts of 6 and 7 later.

Problem 0. Look up the *logistic differential equation* and its sigmoid growth curve in a calculus book. The goal is to understand

- (a) how the logistic differential equation is solved, and
- (b) how what it describes is different from the iterative logistic model.

Problem 1. Study the map

$$x_{n+1} = \sqrt{x_n}.$$

Iterate the map graphically or computationally. Find all fixed points and test their stability. Describe the behavior of the iterates for appropriate choices of the initial condition x_0 .

Problem 2. Study the map

$$x_{n+1} = \tan(x_n).$$

(as for Problem 1).

Problem 3. Try to determine the Feigenbaum constant numerically for the map

$$x_{n+1} = a \sin(x_n).$$

(find successive parameter values at which period-doubling occurs and use these to estimate the Feigenbaum constant).

Problem 4. Assume P_t represents the fraction of neurons of a large neural network that fire at time t . As a simple model of epilepsy, the dynamics of the network can be described by the finite-difference equation

$$P_{t+1} = 4CP_t^3 - 6CP_t^2 + (1 + 2C)P_t,$$

where C is a positive parameter and $0 \leq P_t \leq 1$.

- (a) find the fixed points.
- (b) Determine the stability at each fixed point and describe the dynamics in the neighborhood of the fixed points as a function of C .

- (c) Sketch P_{t+1} as a function of P_t for $C = 4$. Show all maxima, minima, and inflection points.
- (d) Based on what you found, discuss the dynamics as $t \rightarrow \infty$ starting from an initial condition of $P_0 = 0.45$ with $C = 4$. Try to do this graphically and, if possible, on a computer or app or whatever.

Problem 5. Print your last name. Count the number of letters and multiply the number by 0.1. Your *magic number*, m , is 1 plus the number that you just computed. If your last name has nine or more letters, assume that $m = 1.9$.

Consider the finite-difference equation given by the following equations:

$$x_{t+1} = mx_t, \quad 0 \leq x_t \leq \frac{1}{m},$$

$$x_{t+1} = mx_t - 1, \quad \frac{1}{m} < x_t \leq 1.$$

- (a) Draw a graph of x_{t+1} as a function of x_t .
- (b) Determine the fixed point(s) and determine their stability.
- (c) Graphically iterate this equation.
- (d) Are the dynamics chaotic?

Problem 6. Show that if there is one solution to $x_m = f(f(x_m))$ where $x_m \neq f(x_m)$ then there must be another, different solution.

Problem 7. Derive and plot the “3rd iterate” of the logistic map. Look at –do if you like, but beware it could be the beginning of a course project!– problem 10.4.10¹ (The birth of period 3) in Strogatz.

¹2nd edition problem number